Occupancy Modeling

“Abundance of cryptic animals”

Discussion: Occupancy in continuous habitat

Murray G. Efford and Deanna K. Dawson
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Discussion: Occupancy in continuous habitat

Murray G. Efford[1,†] and Deanna K. Dawson[2]

1. Authors suggest that the probability that a plot is occupied depends on three things. What are they and which two are varied in this study?

Plot size, home range size, and population density. Plot size and home range size are varied.

2. Why do the authors say that the concept of plot size is complex?

Plot size is hard to define depending on detection method.

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3. Large plot size vs “point” plots. What are the differing effects on occupancy?

Larger plot size -> chances of detection go down. Point plots make plot size hard to define/very small, so occupancy is more affected by density and home range area.

4. Authors describe occupancy as a “state variable” for animal population monitoring. What might this mean? Can you think of state variables from other disciplines?

State variable = one that describes the system completely enough to determine its future behavior. Eg.) positional coordinates of mechanical parts in mechanical systems.

Discussion:

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5. What is meant by the "temporal dimension of occupancy"?

Is occupancy instantaneous or integrated over time?

6. What is the "closure assumption" referenced in the article?

Closure proposes that sites are closed to changes in occupancy between surveys.

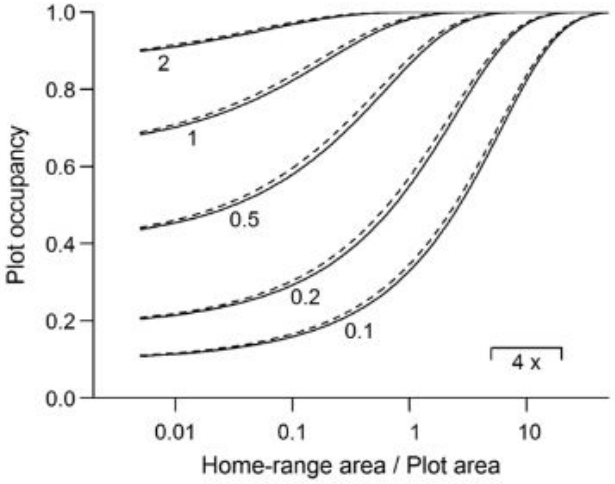
7. Authors say that occupancy is "not exactly" a population parameter. What do you think about that? What IS a population parameter?

Parameter = describes characteristics of an entire population. Occupancy is linked to sampling, so is not a parameter.

So what does each type of plot size do?

Large Plot

- The larger the ratio of home range to plot size, the greater the chance of detection
- Large plots are considered greater than 100 times the area of a home range
- Large plots are the best option for an accurate estimate of population size



So what does each type of plot size do?

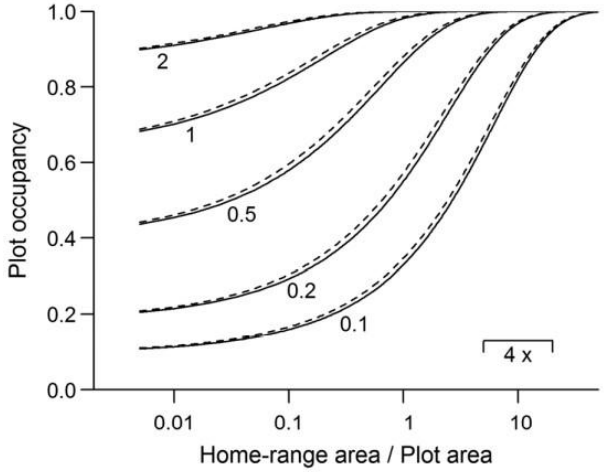
Small Plots

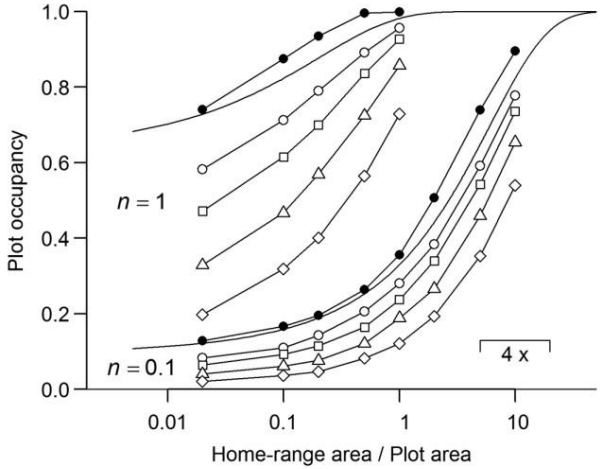
- Point sampling using trail cameras or other techniques
- Assumes that animal ranges have hard boundaries, which has been shown to be inaccurate
- Authors advocate for attention to this underlying assumption in studies that use data from traps or cameras

So what does each type of plot size do?

Intermediately-sized Plots

- Plots are sized closely to home range sizes
- Intermediate sized plots are more prone to issues than larger or smaller plots





Lecture

What is occupancy modeling and how do we do it?

Occupancy Modeling: Overarching Goals

Determine where a species occurs

- Scale dependent!
- Great for cryptic or rare species (mostly utilizes non-invasive techniques)

Identify what factors influence occupancy and detection of a species

- Varies by collection method and species
- Include as covariates in the model

Assumptions

What are the core assumptions of occupancy modeling?

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What are the core assumptions of occupancy modeling?

1. Closure
2. Independence
3. Detection probability (p) is constant or explained by covariates
4. Occupancy probability (ψ) is constant, or explained by covariates

Why use occupancy modeling over other techniques?

Why use occupancy modeling over other techniques?

- **Relatively easy data to collect**
  - Non-invasive
  - Logistically, more feasible (to a point)
- **Many long-term datasets can be adapted to occupancy methods**
  - Monitor presence/absence over long periods of time is more feasible and less expensive
- **Can provide insight on relationships between species and habitat**
- **Large-scale monitoring protocols**
  - Regional monitoring protocols can be implemented to collect data used for occupancy modeling
- **Can sometimes use occupancy as a proxy for abundance**

Occupancy-Abundance Relationship

When might we be able to use occupancy as a proxy for abundance?

Occupancy-Abundance Relationship

When might we be able to use occupancy as a proxy for abundance?

This is quite tricky!

- We need to understand what might *drive* the shape of a positive relationship between occupancy and abundance
  - Strictly mathematical?
  - Ecological factors (density, HR size, etc.)
  - Our understanding of the definition of abundance (how it is calculated/defined varies)
  - Scale (temporal and spatial)

Occupancy-Abundance Relationship

We might assume a proxy of occupancy for abundance when:

- Occupancy is estimated in discrete habitat patches
- Organisms are either immobile or have limited mobility
- We can combine methods with mark-recapture/individual identification
- Density is low/little to no overlap between individuals (territoriality helps!)

Take-away = *Use caution if you try to infer about abundance from occupancy models. There are a lot of factors that drive the shape and magnitude of this relationship.*

Data collection methods and their assumptions

Collection Strategies

- Camera traps
- Transect surveys
  - Sign (tracks, scat, etc.)
  - Point counts (bird song)
  - Coverboard arrays
- Aerial surveys
  - Flown typically in transects

How might our assumptions of closure and independence be violated?

Basic Occupancy Model

y: the data at site i

p: detection probability

z: true occupancy state at site i

ψ: occupancy probability

α: the parameters to estimate for the detection probability p

β: the parameters to estimate for the occupancy probability ψ

$$y_{i,j} \sim Bernoulli(p_{i,j}) * z_i$$

$$z_i \sim Bernoulli(\psi_i)$$

$$\psi_i = \beta * \mathbf{x_i^1}$$

$$p_{i,j} = \alpha * \mathbf{x_{i,j}^2}$$

Uses and limitations

- Probability of detection (p) can be a nuisance variable or provide valuable insight, takes some thinking about why a species may or may not be detectable in the system
- State variable (occupancy) usually not as interesting to managers as abundance

Basic Occupancy Simulations

‘`unmarked`’ and `JAGS`

Simulating Occupancy

```r
library(unmarked)
n.site=100
x1=rnorm(n.site)
x2=rnorm(n.site)
covs=data.frame(x1,x2)
beta=list("beta0"=0.5,"beta1"=-2,"beta2"=1)
alpha=list("alpha0"=1,"alpha1"=1)
logit.psi=beta$beta0+beta$beta1*covs$x1+beta$beta2*covs$x2
expit=function(x){exp(x)/(1+exp(x))}
psi=expit(logit.psi)
z=rbinom(n = n.site,size = 1,prob = psi)
logit.p=alpha$alpha0+alpha$alpha1*covs$x1
p.uncond=expit(logit.p)
p=p.uncond*z
y1=rbinom(n=n.site,size=1,prob=p)
y2=rbinom(n=n.site,size=1,prob=p)
y3=rbinom(n=n.site,size=1,prob=p)
y=cbind(y1,y2,y3)
###Unmarked Method
umf=unmarkedFrameOccu(y=y,siteCovs = covs)
occu(formula = ~x1 ~x1+x2,data = umf)
```

```
Occupancy:
            Estimate    SE     z P(>|z|)
(Intercept)    0.882 0.371  2.37  0.0176
x1            -1.579 0.619 -2.55  0.0107
x2             0.518 0.304  1.71  0.0880

Detection:
            Estimate    SE    z  P(>|z|)
(Intercept)    0.975 0.215 4.54 5.62e-06
x1             1.024 0.258 3.96 7.34e-05
```

Simulating Occupancy

```
beta=list("beta0"=0.5,"beta1"=-2,"beta2"=1)
alpha=list("alpha0"=1,"alpha1"=1)
```

```
n.sim=1000
coeff.ests=matrix(NA,nrow=n.sim,ncol=5)
for(i in 1:n.sim){
z=rbinom(n = n.site,size = 1,prob = psi)
p=p.uncond*z
y1=rbinom(n=n.site,size=1,prob=p)
y2=rbinom(n=n.site,size=1,prob=p)
y3=rbinom(n=n.site,size=1,prob=p)
y=cbind(y1,y2,y3)
###Unmarked Method
umf=unmarkedFrameOccu(y=y,siteCovs = covs)
um=occu(formula = ~x1 ~x1+x2,data = umf)
coeff.ests[i,]=coef(um)
}
```

```
mean.est=apply(coeff.ests,2,mean)
names(mean.est)=c("psi(Int)","psi(x1)","psi(x2)","p(Int)","p(x1)"
mean.est
```

```
##   psi(Int)    psi(x1)    psi(x2)     p(Int)      p(x1)
##  0.5449195 -2.1889175  1.0928729  1.0240093  1.0330155
```

Simulating Occupancy with JAGS

```
###Jags Method
model_text="model
{

# Likelihood
for (m in 1:M){ # Model for occurrence at site level
z[m] ~ dbern(psi[m])
logit(psi[m])=beta0+beta1*x1[m]+beta2*x2[m]

for (i in 1:P){ # Observation model at observation level
y[m,i] ~ dbern(p.eff[m,i])
p.eff[m,i]=p[m,i]*z[m]
logit(p[m,i]) <- alpha0+alpha1*x1[m]
}}
# Priors
beta0~dlogis(0,1) # Occupancy intercept
beta1~dlogis(0,1)
beta2~dlogis(0,1)
alpha0~dlogis(0,1)
alpha1~dnorm(0, 0.3)
}
"
inits=function(){list(z=apply(y, 1, max,na.rm=T),
                      alpha0=runif(1),alpha1=runif(1),
                      beta0=runif(1),beta1=runif(1),beta2=runif(1))}
JM=run.jags(model=model_text,
            monitor = c("alpha0","alpha1",
                        "beta0","beta1","beta2"),
            data = list(M=n.site,P=3,y=y,x1=covs$x1,x2=covs$x2),
            burnin = 10000,n.chains = 4,sample = 20000,method = "rjags",inits=inits)
```

Simulating Occupancy with JAGS

```r
beta=list("beta0"=0.5,"beta1"=-2,"beta2"=1)
alpha=list("alpha0"=1,"alpha1"=1)
```

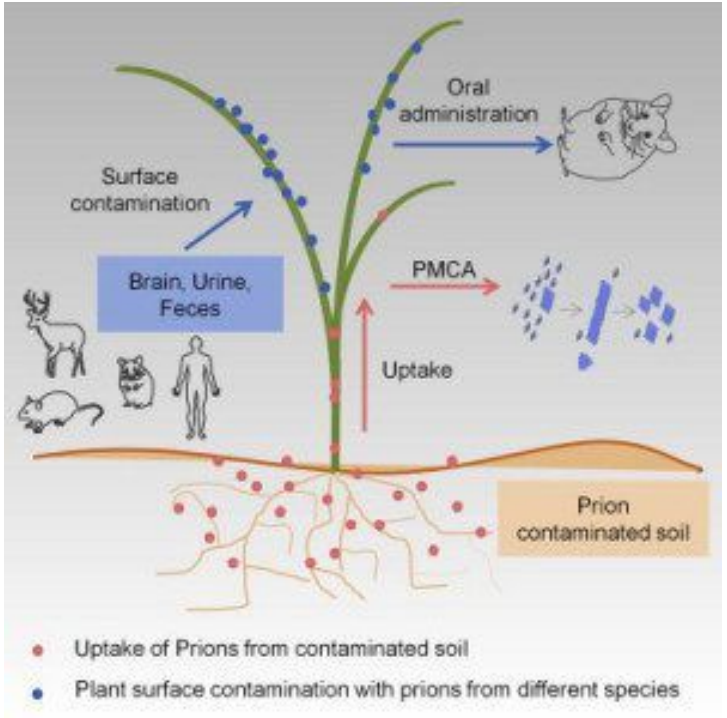
```
> summary(JM)[,2]
    alpha0     alpha1      beta0      beta1      beta2
 0.9309903  1.0176299  0.1858387 -1.8843910  0.7995822
```

Case Study: Scrapie

Prevalence of prions on the landscape

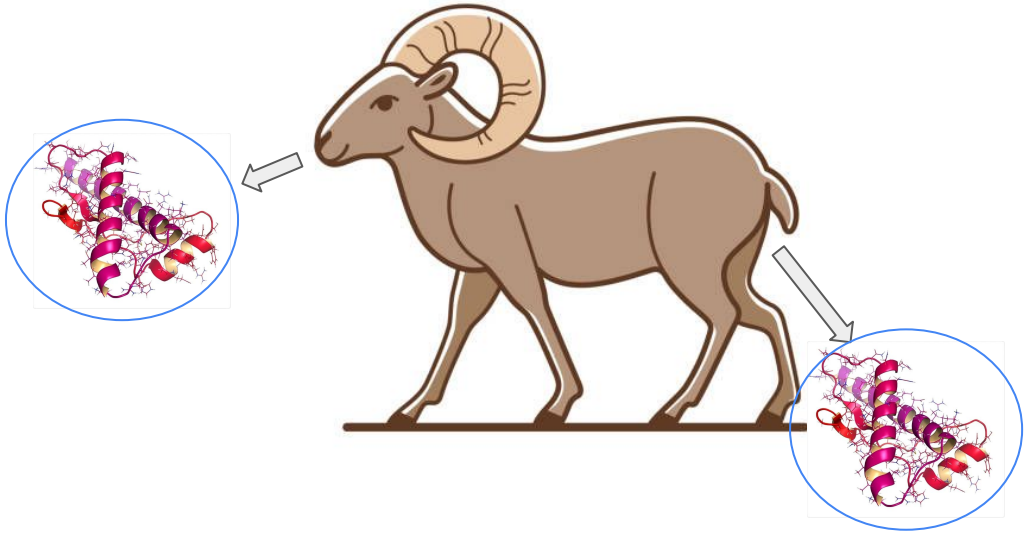
Background

Scrapie is a disease caused by a prion (similar to CWD)



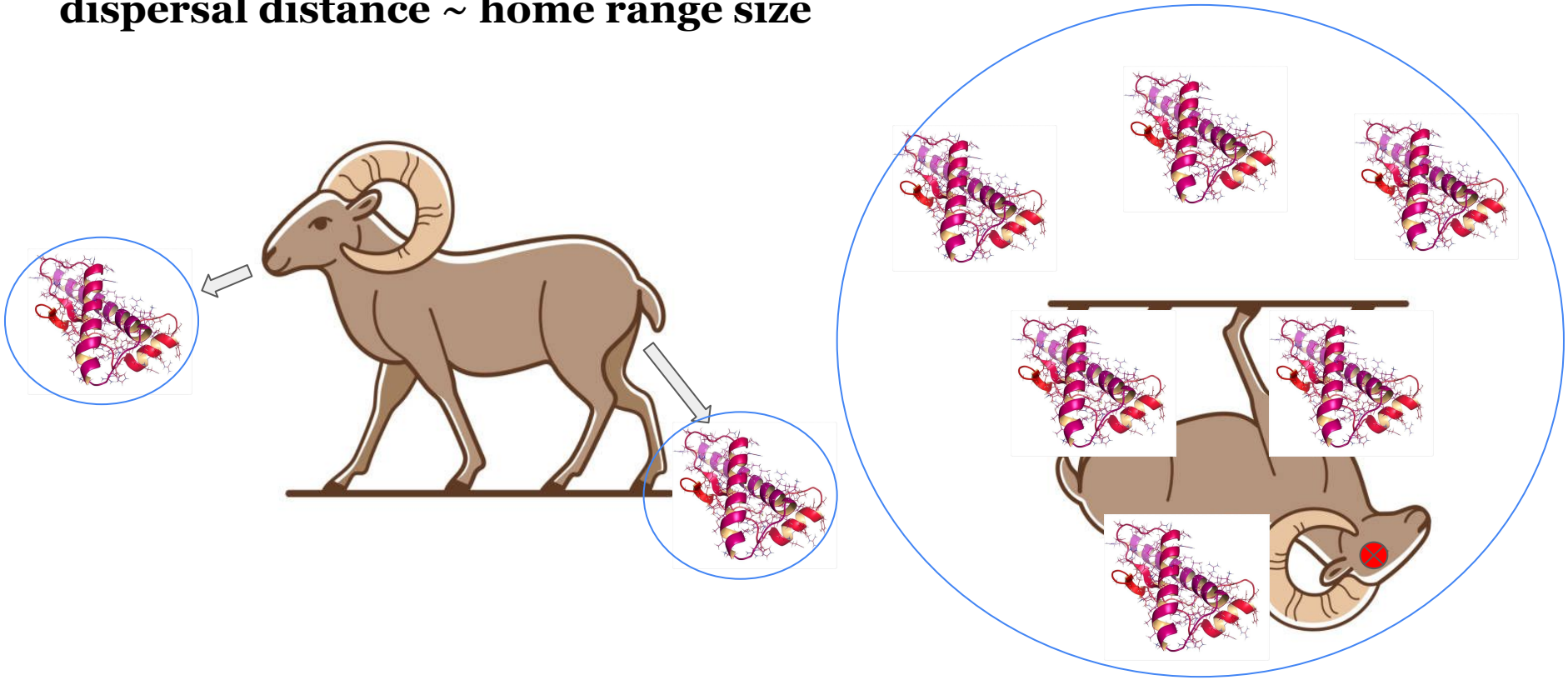
Courtesy of Dr. Michelle L. Crocheck, USDA-APHIS-VS-NVSL

Limited Knowledge of Prion Dispersal Distances



Limited Knowledge of Prion Dispersal Distances

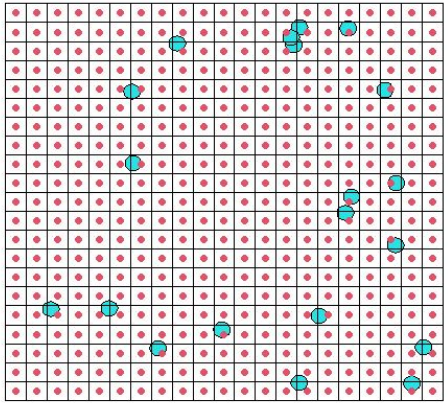
dispersal distance ~ home range size



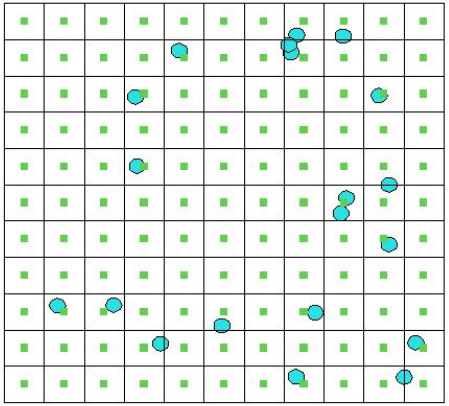
Goals

- **How do the following affect occupancy (prevalence) estimates:**
  - Spatial scale - sampling grid
    - $5m^2$ vs $10m^2$
  - Dispersal distance (home range) size
    - $2m^2$ vs $15m^2$
  - Intensity (scattered vs. concentrated)
    - In-class = scattered/less prevalent

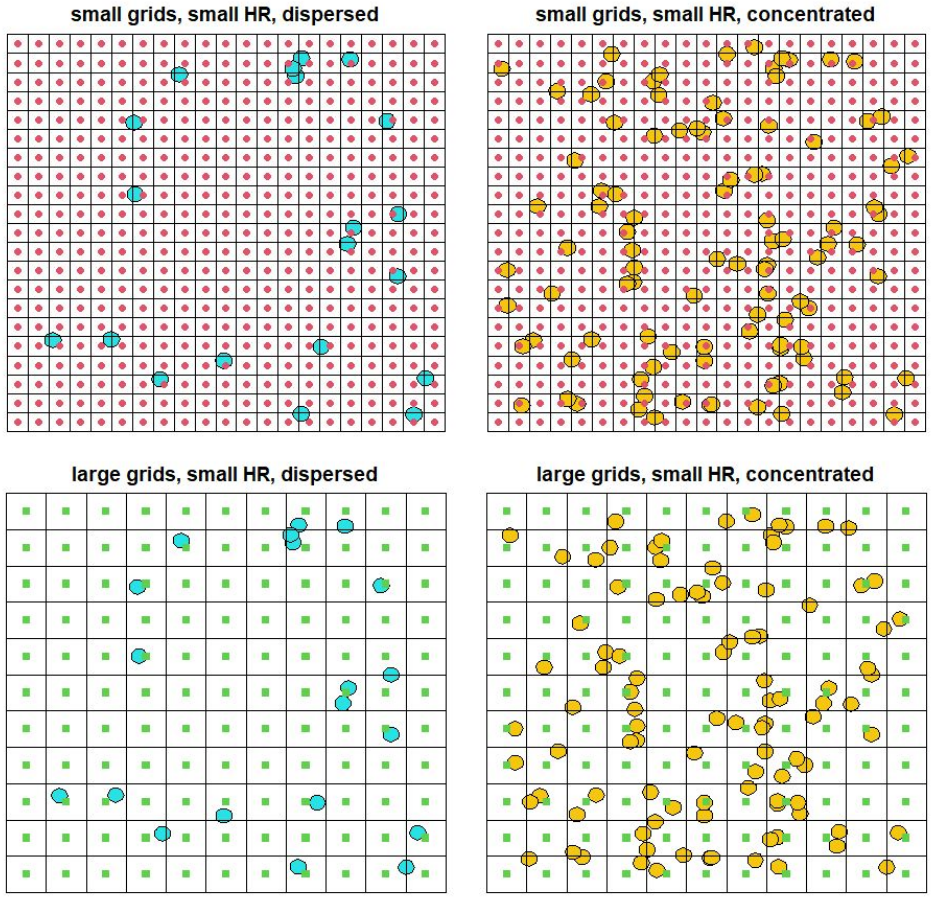
small grids, small HR, dispersed



large grids, small HR, dispersed

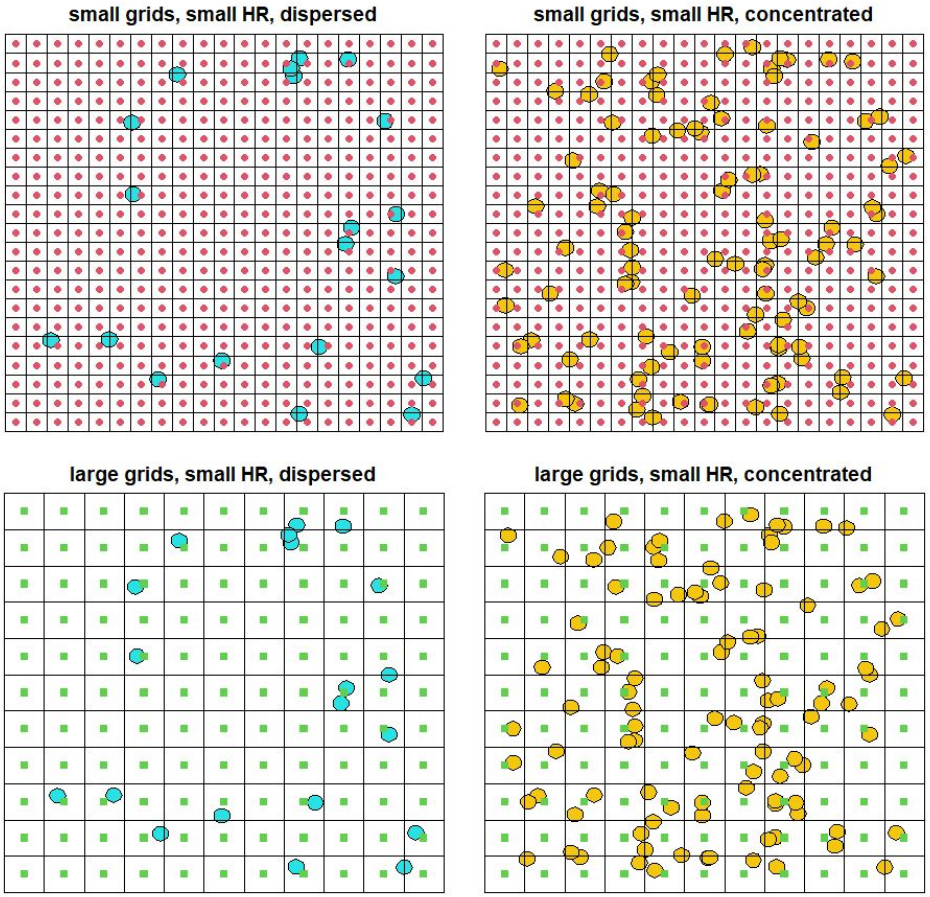


rare species/
low intensity outbreak



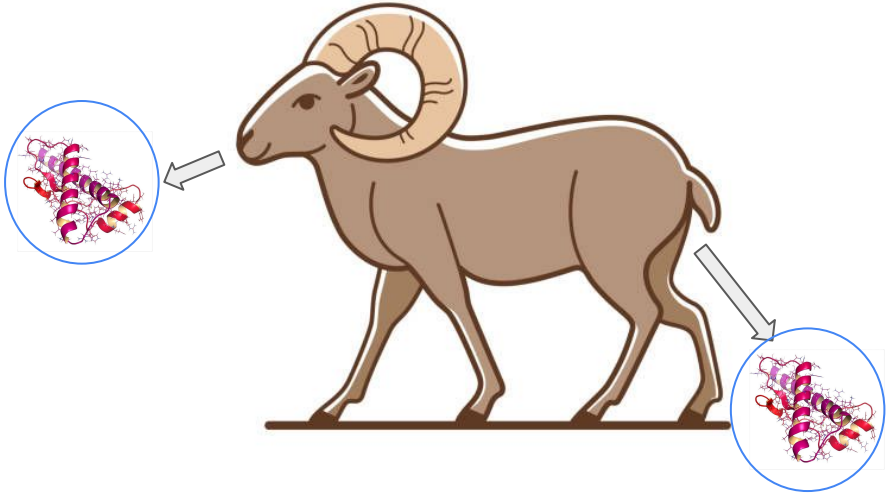
rare species/
low intensity outbreak

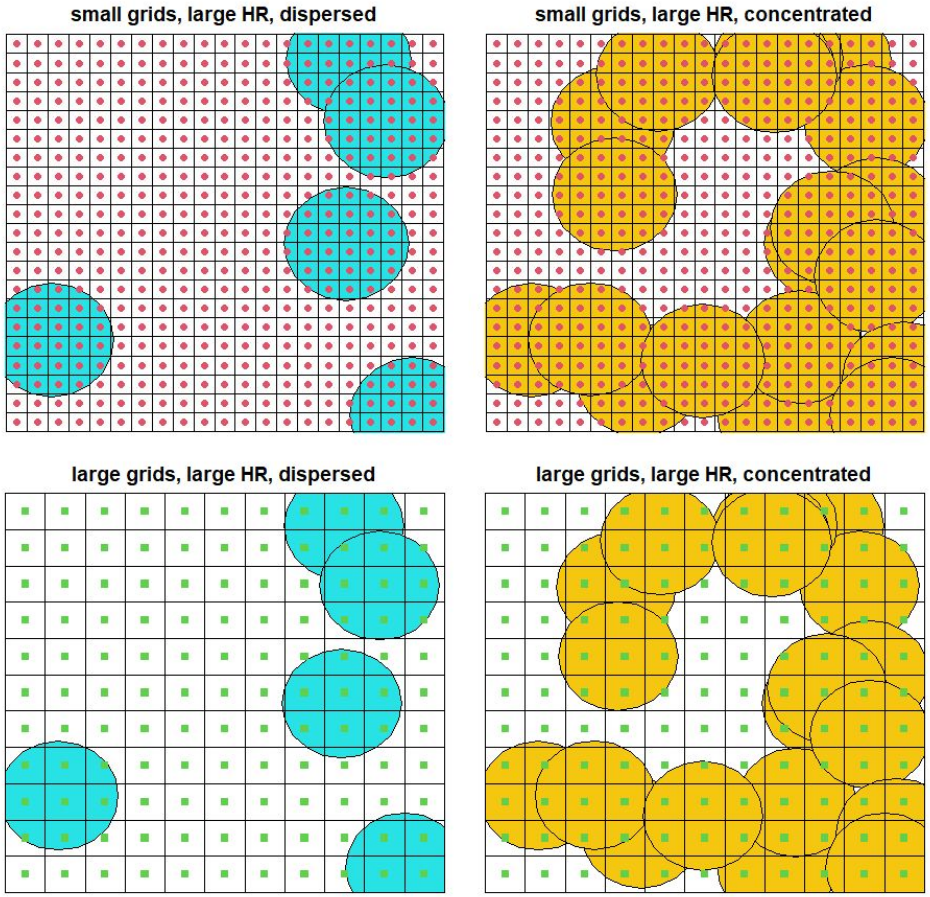
common species/
high intensity outbreak



rare species/
low intensity outbreak

common species/
high intensity outbreak





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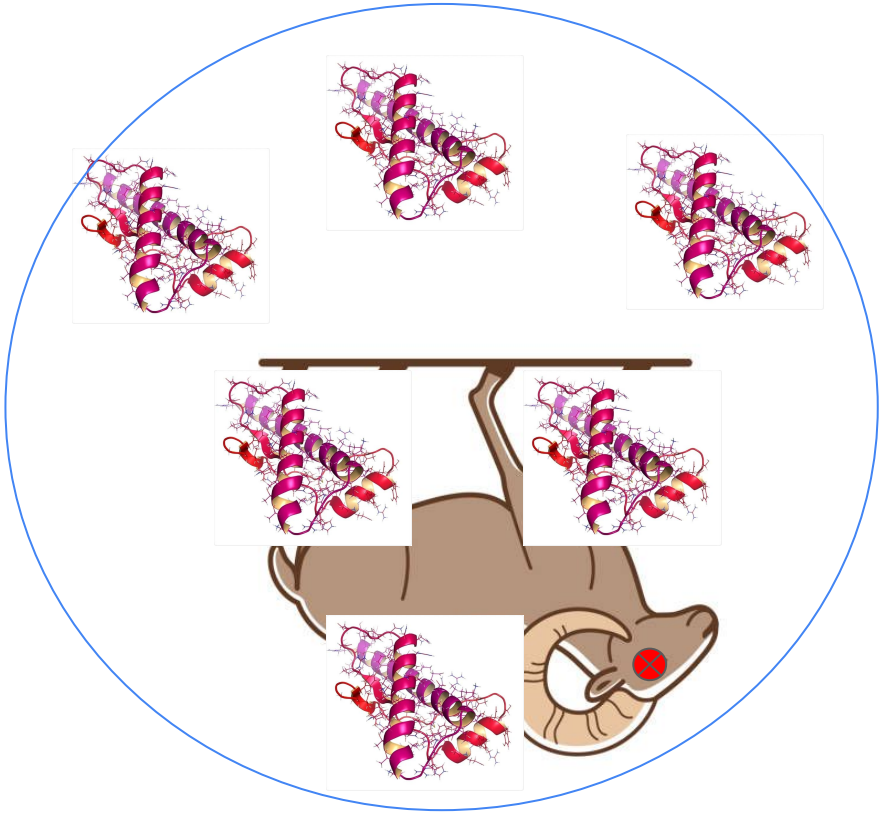
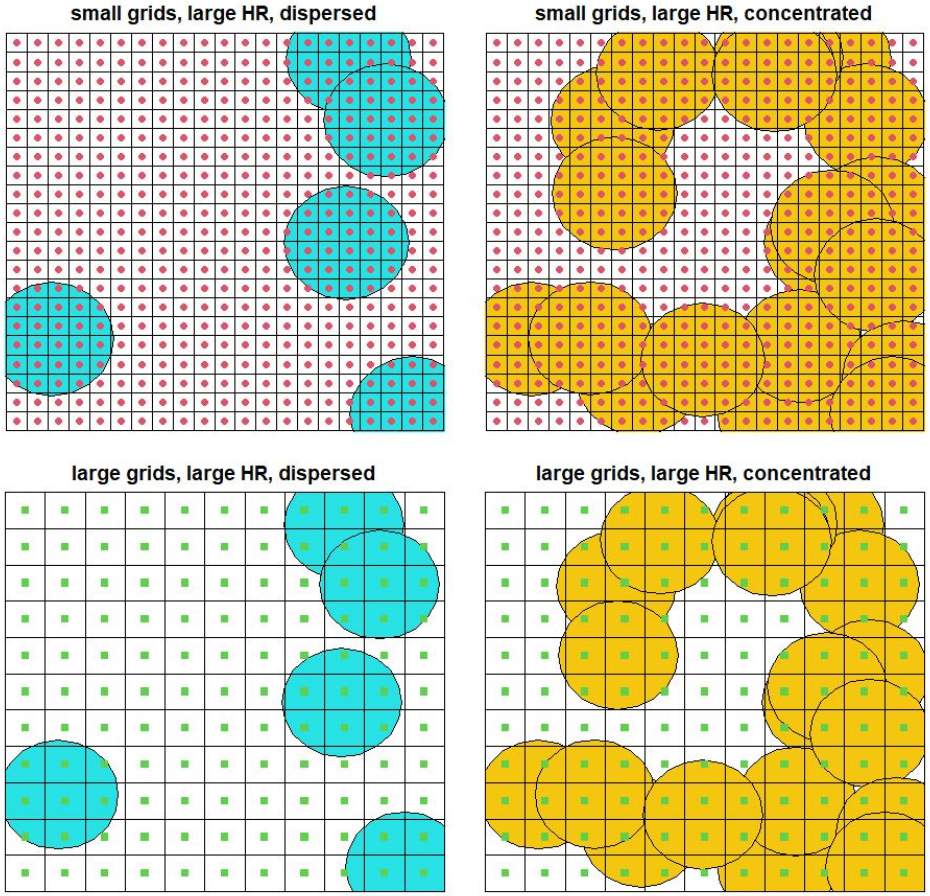
low intensity outbreak

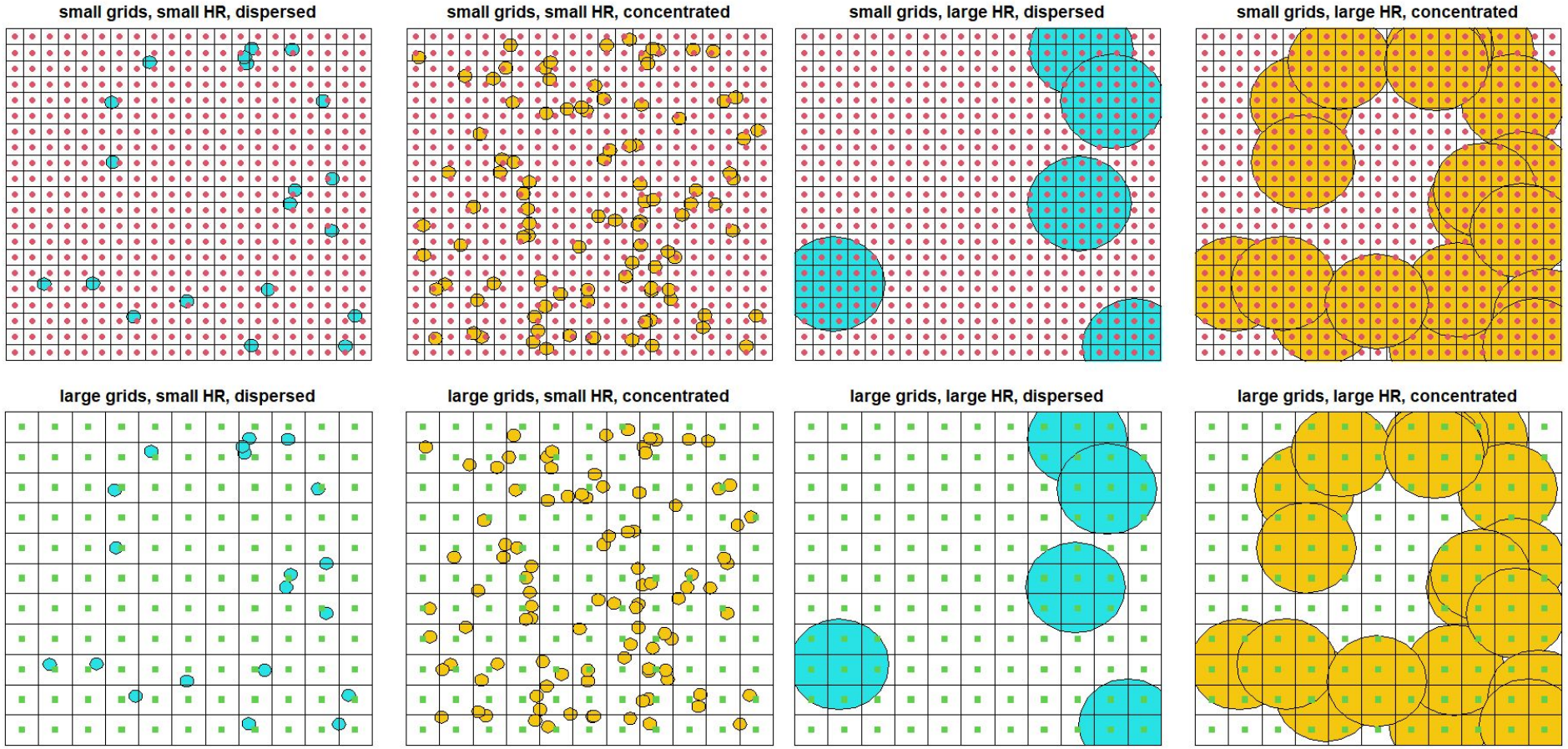
common species/
high intensity outbreak





rare species/
low intensity outbreak

common species/
high intensity outbreak



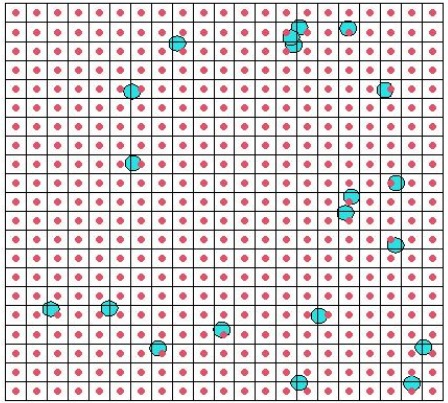
rare species/ low intensity outbreak

common species/ high intensity outbreak

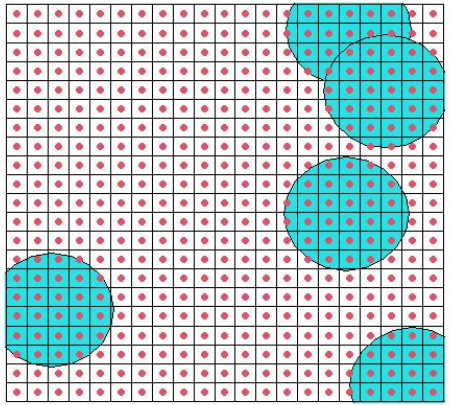
rare species/ low intensity outbreak

common species/ high intensity outbreak

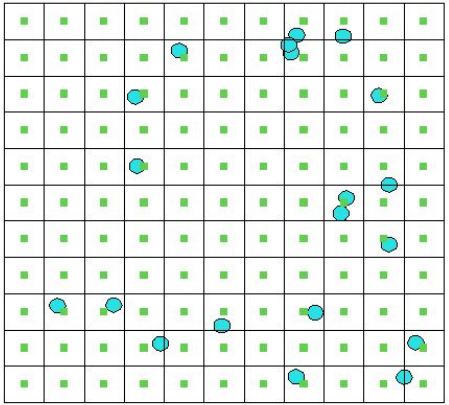
small grids, small HR, dispersed

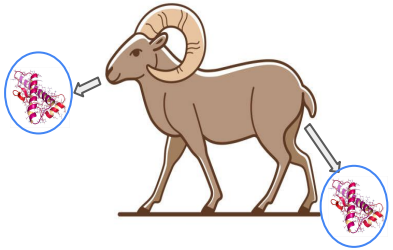


small grids, large HR, dispersed

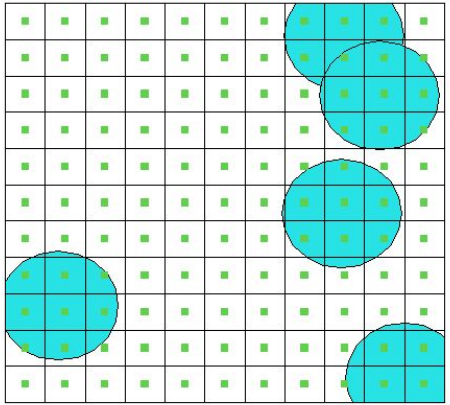


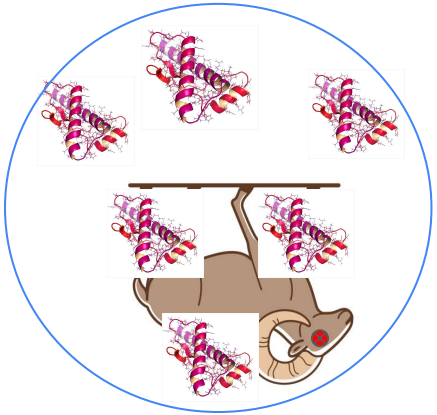
large grids, small HR, dispersed





large grids, large HR, dispersed





rare species/
low intensity outbreak

rare species/
low intensity outbreak

Homework

Simulate your own occupancy study for a disease causing prion!

Create a sampling grid like we just did in class *but*:

- Consider it a common species (concentrated instead of dispersed)
- Run both a large and small home range
- Use small and large grid sizes
- Compare these scenarios as well as what we've seen in class
- What type of sampling should we use to most effectively understand this outbreak?